

Code

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*&-----*
*& Report Z_DTTS_REPROCESS_COCKPIT
*&-----*
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*&-----*
REPORT Z_DTTS_REPROCESS_COCKPIT.

TABLES: mseg.

*" Variables for dynamic proxy-based web service call (config-driven via ZMM_DTTS_API_
CON)
DATA: request_ptr      TYPE REF TO data,          "Dynamic request structure po
inter
      response_ptr     TYPE REF TO data,          "Dynamic response structure p
ointer
      lo_proxy         TYPE REF TO object,        "Web service proxy object
      p_lport          TYPE prx_logical_port_name, "Logical port name for proxy
      ptab             TYPE abap_parmbind_tab,    "Parameter binding table for
proxy
      wa_ptab          TYPE abap_parmbind,        "Single parameter binding ent
ry
      wa_zmm_dtts_api_con TYPE zmm_dtts_api_con. "API config row (proxy class,
method etc.)
TYPES: BEGIN OF ty_alv_data,
      selkz           TYPE char1,
      "mandt          TYPE mandt,
      tran_id         TYPE n LENGTH 20,
      item_no         TYPE numc4,
      zeile           TYPE mblpo,
      product         TYPE matnr,
      prod_name       TYPE maktx,
      prod_qty        TYPE menge_d,
      prod_unit       TYPE meins,
      gtin            TYPE z_dgtin,
      batch           TYPE charg_d,
      exp_date        TYPE vfdat,
      notif_id        TYPE char50,
      tr_response     TYPE string,
      mat_doc         TYPE mblnr,
      mvt_type        TYPE bwart,
      sr_number       TYPE char50,
      created_date    TYPE dats,
      created_time    TYPE tims,
      created_by      TYPE xubname,
      changed_date    TYPE dats,
      changed_time    TYPE tims,
      changed_by      TYPE xubname,
      prod_stat       TYPE char10,
      trans_stat      TYPE char10,
      doc_year        TYPE mjahr,
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        item_operation    TYPE char20,
        header_operation TYPE char20,
        frm_gln           TYPE zmm_sst_gln_src,
        to_gln            TYPE zmm_sst_gln_des,
        auth_gln          TYPE zmm_sst_gln_auth,
    END OF ty_alv_data.
DATA: gt_alv_data TYPE TABLE OF ty_alv_data.
*-----*
* SELECTION SCREEN
*-----*
SELECTION-SCREEN BEGIN OF BLOCK b1 WITH FRAME TITLE TEXT-001.
    SELECT-OPTIONS: s_matdoc FOR mseg-mblnr,
                    s_year   FOR mseg-mjahr,
                    s_bwart  FOR mseg-bwart.
    PARAMETERS: p_oper TYPE char20.
SELECTION-SCREEN END OF BLOCK b1.

*-----*
* LOCAL CLASS DEFINITION
*-----*
CLASS lcl_dtts_cockpit DEFINITION.
    PUBLIC SECTION.
        TYPES: BEGIN OF ty_alv_data,
            selkz          TYPE char1,
            "mandt         TYPE mandt,
            tran_id        TYPE ztran_id,
            item_no        TYPE numc4,
            zeile          TYPE mblpo,
            product        TYPE matnr,
            prod_name      TYPE maktx,
            prod_qty       TYPE menge_d,
            prod_unit      TYPE meins,
            gtin           TYPE z_dgtin,
            batch          TYPE charg_d,
            exp_date       TYPE vfdat,
            notif_id       TYPE char50,
            tr_response    TYPE string,
            mat_doc        TYPE mblnr,
            mvt_type       TYPE bwart,
            sr_number      TYPE char50,
            created_date   TYPE dats,
            created_time   TYPE tims,
            created_by     TYPE xubname,
            changed_date   TYPE dats,
            changed_time   TYPE tims,
            changed_by     TYPE xubname,
            prod_stat      TYPE char10,
            trans_stat     TYPE char10,
            doc_year       TYPE mjahr,
            item_operation TYPE char20,
            header_operation TYPE char20,
            frm_gln        TYPE zmm_sst_gln_src,
            to_gln         TYPE zmm_sst_gln_des,
            auth_gln       TYPE zmm_sst_gln_auth,
        END OF ty_alv_data.

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DATA: gt_alv_data TYPE TABLE OF ty_alv_data.

DATA: go_alv          TYPE REF TO cl_gui_alv_grid,
      go_container TYPE REF TO cl_gui_custom_container.

METHODS:
  get_data,
  display_alv,
  handle_user_command FOR EVENT user_command OF cl_gui_alv_grid
    IMPORTING e_ucomm,
  handle_toolbar FOR EVENT toolbar OF cl_gui_alv_grid
    IMPORTING e_object e_interactive.

PRIVATE SECTION.
  METHODS:
    reprocess_selected_items.
ENDCLASS.

*-----*
* LOCAL CLASS IMPLEMENTATION
*-----*
CLASS lcl_dtts_cockpit IMPLEMENTATION.

  METHOD get_data.
    " Retrieve data based on selection screen
    " ZIMMDTTS_4 is a view mapping to ZIMM_DTTS_FY_HELPER left outer join zmm_sst_dtts
    _hdr
    SELECT ""itm~mandt,
           itm~tran_id,
           itm~item_no,
           itm~zeile,
           itm~product,
           itm~prod_name,
           itm~prod_qty,
           itm~prod_unit,
           itm~gtin,
           itm~batch,
           itm~exp_date,
           itm~notif_id,
           itm~tr_response,
           itm~mat_doc,
           itm~mvt_type,
           itm~sr_number,
           itm~created_date,
           itm~created_time,
           itm~created_by,
           itm~changed_date,
           itm~changed_time,
           itm~changed_by,
           itm~prod_stat,
           itm~trans_stat,
           itm~doc_year,
           itm~item_operation,
           hdr~operation AS header_operation,
           hdr~frm_gln,
           hdr~to_gln,

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        hdr~auth_gln
FROM zimm_dttts_4 AS itm
LEFT OUTER JOIN zmm_sst_dttts_hdr AS hdr ON itm~mat_doc = hdr~mat_doc
                                         AND itm~doc_year = hdr~doc_yr
                                         AND itm~mvt_type = hdr~mvt_type

WHERE itm~mat_doc IN @s_matdoc
      AND itm~doc_year IN @s_year
      AND itm~mvt_type IN @s_bwart
      ""AND hdr~operation = @p_oper
INTO CORRESPONDING FIELDS OF TABLE @gt_alv_data.

IF sy-subrc <> 0.
    MESSAGE 'No data found for the given selection criteria.' TYPE 'S' DISPLAY LIKE
'E'.
ENDIF.
ENDMETHOD.

METHOD display_alv.
    DATA: lt_fcat TYPE lvc_t_fcat,
           ls_layout TYPE lvc_s_layo.

    IF go_container IS INITIAL.
        " Need to create a screen 0100 for this ALV or use full screen CL_SALV_TABLE.
        " For simplicity in full screen without screen painter, we use cl_gui_alv_grid w
ith default full screen container.
        * CREATE OBJECT go_alv
        * EXPORTING
        * i_parent = cl_gui_container=>screen0.
        CREATE OBJECT go_container
        EXPORTING
            container_name = 'CC_ALV'.

        CREATE OBJECT go_alv
        EXPORTING
            i_parent = go_container.

        " Prepare layout
        ls_layout-sel_mode = 'A'.
        ls_layout-box_fname = 'SELKZ'.

        " Build Field Catalog dynamically or manually
        CALL FUNCTION 'LVC_FIELDCATALOG_MERGE'
        EXPORTING
            i_structure_name = 'ZIMM_DTTTS_FY_HELPER' " Approximation, manual setup is be
tter for custom fields
        CHANGING
            ct_fieldcat = lt_fcat
        EXCEPTIONS
            OTHERS = 1.
        DATA ls_fcat LIKE LINE OF lt_fcat.

        CLEAR ls_fcat.
        ls_fcat-fieldname = 'SELKZ'.
        ls_fcat-checkbox = 'X'.
        ls_fcat-edit = 'X'.

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APPEND ls_fcat TO lt_fcat.
CLEAR ls_fcat.

" Mark fields as editable (allow user to edit data in ALV directly)
LOOP AT lt_fcat ASSIGNING FIELD-SYMBOL(<fs_fcat>).
    <fs_fcat>-edit = 'X'.
ENDLOOP.

" Set Handlers
SET HANDLER me->handle_toolbar FOR go_alv.
SET HANDLER me->handle_user_command FOR go_alv.

" Display ALV
go_alv->set_table_for_first_display(
    EXPORTING
        is_layout      = ls_layout
    CHANGING
        it_outtab      = gt_alv_data
        it_fieldcatalog = lt_fcat ).
ELSE.
    go_alv->refresh_table_display( ).
ENDIF.
ENDMETHOD.

METHOD handle_toolbar.
    DATA: ls_toolbar TYPE stb_button.
    CLEAR ls_toolbar.
    ls_toolbar-function = 'REPROCESS'.
    ls_toolbar-icon = '@5C@'. " Reprocess Icon
    ls_toolbar-quickinfo = 'Reprocess Selected Items'.
    ls_toolbar-butn_type = 0.
    ls_toolbar-text = 'Reprocess'.
    APPEND ls_toolbar TO e_object->mt_toolbar.
ENDMETHOD.

METHOD handle_user_command.
    CASE e_ucomm.
        WHEN 'REPROCESS'.
            go_alv->check_changed_data( ).
            me->reprocess_selected_items( ).
    ENDCASE.
ENDMETHOD.

METHOD reprocess_selected_items.
    DATA: lt_selected      TYPE TABLE OF ty_alv_data,
          lt_unique        TYPE TABLE OF ty_alv_data,
          lt_header        TYPE TABLE OF zmm_sst_dtts_hdr,
          lt_items         TYPE TABLE OF zmm_sst_dtts_itm,
          lt_mseg          TYPE TABLE OF mseg,
          wa_zmm_dtts_api_con TYPE zmm_dtts_api_con,
          ptab            TYPE abap_parmbind_tab,
          wa_ptab          LIKE LINE OF ptab,
          p_lport         TYPE prx_logical_port_name,
          lo_proxy         TYPE REF TO object.

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" 1. Extract selected lines
LOOP AT gt_alv_data INTO DATA(ls_alv) WHERE selkz = 'X'.
  APPEND ls_alv TO lt_selected.
ENDLOOP.

IF lt_selected IS INITIAL.
  MESSAGE 'Please select at least one item to reprocess.' TYPE 'I'.
  RETURN.
ENDIF.

" 2. Remove items that have status SUCCESS
DELETE lt_selected WHERE prod_stat = 'SUCCESS'.

IF lt_selected IS INITIAL.
  MESSAGE 'No eligible items to reprocess (all selected were SUCCESS).' TYPE 'I'.
  RETURN.
ENDIF.

" 3. Sort selected items using key matdoc, mjahr and movement type
SORT lt_selected BY mat_doc doc_year mvt_type.

" 4. Create a new table with only unique matdoc, mjahr and movement type
lt_unique = lt_selected.
DELETE ADJACENT DUPLICATES FROM lt_unique COMPARING mat_doc doc_year mvt_type.

" 5. Loop in this unique table
LOOP AT lt_unique INTO DATA(ls_unique).
  CLEAR: lt_header, lt_items.

  " Get the header row from header table and append it
  SELECT * FROM zmm_sst_dtts_hdr
    INTO TABLE @DATA(lt_hdr_tmp)
    WHERE mat_doc = @ls_unique-mat_doc
      AND doc_yr = @ls_unique-doc_year
      AND mvt_type = @ls_unique-mvt_type.
  APPEND LINES OF lt_hdr_tmp TO lt_header.

  DATA(lv_operation) = ls_unique-header_operation.

  " Loop at the selected items table for the item lines and append to lt_items
  LOOP AT lt_selected INTO DATA(ls_sel_item)
    WHERE mat_doc = ls_unique-mat_doc
      AND doc_year = ls_unique-doc_year
      AND mvt_type = ls_unique-mvt_type.

    DATA: ls_item TYPE zmm_sst_dtts_itm.
    MOVE-CORRESPONDING ls_sel_item TO ls_item.
    APPEND ls_item TO lt_items.
  ENDLOOP.

  " =====
  " LOAD API CONFIGURATION FROM ZMM_DTTS_API_CON
  " =====
  SELECT SINGLE * FROM zmm_dtts_api_con INTO @wa_zmm_dtts_api_con
    WHERE api_name = @lv_operation.

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IF sy-subrc = 0.
    " Dynamically create request and response objects based on config
    CREATE DATA request_ptr TYPE (wa_zmm_dtts_api_con-request_structure).
    ASSIGN request_ptr->* TO FIELD-SYMBOL(<fs_request>).

    CREATE DATA response_ptr TYPE (wa_zmm_dtts_api_con-response_structure).
    ASSIGN response_ptr->* TO FIELD-SYMBOL(<fs_response>).

    " =====
    " BUILD API REQUEST BASED ON OPERATION TYPE
    " =====
    CASE lv_operation.
        WHEN 'ACCEPT'.
            PERFORM accept_request USING lt_mseg lt_header lt_items CHANGING <fs_request>.
        WHEN 'RETURN'.
            PERFORM return_request USING lt_mseg lt_header lt_items CHANGING <fs_request>.
        WHEN 'DISPATCH'.
            PERFORM dispatch_request USING lt_mseg lt_header lt_items CHANGING <fs_request>.
        WHEN 'DISPATCH_CANCEL'.
            PERFORM dispatch_cancel_request USING lt_mseg lt_header lt_items CHANGING <fs_request>.
        WHEN 'TRANSFER'.
            PERFORM transfer_request USING lt_mseg lt_header lt_items CHANGING <fs_request>.
        WHEN 'TRANSFER_CANCEL'.
            PERFORM transfer_cancel_request USING lt_mseg lt_header lt_items CHANGING <fs_request>.
        WHEN 'CONSUME'.
            PERFORM consume_request USING lt_mseg lt_header lt_items CHANGING <fs_request>.
        WHEN 'CONSUME_CANCEL'.
            PERFORM consume_cancel_request USING lt_mseg lt_header lt_items CHANGING <fs_request>.
        WHEN 'DRUG_SALE'.
            PERFORM pharmacy_sale_request USING lt_mseg lt_header lt_items CHANGING <fs_request>.
        WHEN 'DRUG_SALE_CANCEL'.
            PERFORM pharmacy_sale_cancel_request USING lt_mseg lt_header lt_items CHANGING <fs_request>.
        WHEN 'DEACTIVATE'.
            PERFORM deactivate_request USING lt_mseg lt_header lt_items CHANGING <fs_request>.
        WHEN 'DEACTIVATE_CANCEL'.
            PERFORM deactivate_cancel_request USING lt_mseg lt_header lt_items CHANGING <fs_request>.
    ENDCASE.

    " =====
    " CREATE PROXY AND CALL DTTS WEB SERVICE
    " =====

    CLEAR ptab.
    CLEAR wa_ptab.
    wa_ptab-name = 'LOGICAL_PORT_NAME'.

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wa_ptab-kind = cl_abap_objectdescr=>exporting.
p_lport = wa_zmm_dtts_api_con-logical_port.
wa_ptab-value = REF #( p_lport ).
INSERT wa_ptab INTO TABLE ptab.

TRY.
  CREATE OBJECT lo_proxy TYPE (wa_zmm_dtts_api_con-proxy_class)
    PARAMETER-TABLE ptab.

  CALL METHOD lo_proxy->(wa_zmm_dtts_api_con-method_name)
    EXPORTING
      input = <fs_request>
    IMPORTING
      output = <fs_response>.

  " =====
  " PROCESS API RESPONSE AND UPDATE Z-TABLE RECORDS
  " =====
  CASE lv_operation.
    WHEN 'ACCEPT'.
      PERFORM accept_response USING <fs_response> CHANGING lt_header lt_it
ems.
    WHEN 'RETURN'.
      PERFORM return_response USING <fs_response> CHANGING lt_header lt_it
ems.
    WHEN 'DISPATCH'.
      PERFORM dispatch_response USING <fs_response> CHANGING lt_header lt_
items.
    WHEN 'DISPATCH_CANCEL'.
      PERFORM dispatch_cancel_response USING <fs_response> CHANGING lt_he
ader lt_items.
    WHEN 'TRANSFER'.
      PERFORM transfer_response USING <fs_response> CHANGING lt_header lt_
items.
    WHEN 'TRANSFER_CANCEL'.
      PERFORM transfer_cancel_response USING <fs_response> CHANGING lt_he
ader lt_items.
    WHEN 'CONSUME'.
      PERFORM consume_response USING <fs_response> CHANGING lt_header lt_i
tems.
    WHEN 'CONSUME_CANCEL'.
      PERFORM consume_cancel_response USING <fs_response> CHANGING lt_head
er lt_items.
    WHEN 'DRUG_SALE'.
      PERFORM pharmacy_sale_response USING <fs_response> CHANGING lt_heade
r lt_items.
    WHEN 'DRUG_SALE_CANCEL'.
      PERFORM pharmacy_sale_cancel_response USING <fs_response> CHANGING l
t_header lt_items.
    WHEN 'DEACTIVATE'.
      PERFORM deactivate_response USING <fs_response> CHANGING lt_header l
t_items.
    WHEN 'DEACTIVATE_CANCEL'.
      PERFORM deactivate_cancel_response USING <fs_response> CHANGING lt_h
eader lt_items.
  ENDCASE.

```



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" Custom Table updates for reprocess
IF lt_header IS NOT INITIAL.
    MODIFY zmm_sst_dtts_hdr FROM TABLE lt_header.
ENDIF.

" Instead of ZMM_SST_DTTS_ITM, we append a new line to ZMM_SST_DTTSIT2 if
it does not exist,
" else modify the status alone. The new line will have same data as old ta
ble except PROD_STAT and Trans Status.
LOOP AT lt_items INTO DATA(ls_new_item).
    DATA: ls_dttsit2 TYPE zmm_sst_dttsit2.

    " Check if line exists
    SELECT SINGLE * FROM zmm_sst_dttsit2 INTO @ls_dttsit2
    "WHERE mandt    = @ls_new_item-mandt
    WHERE tran_id = @ls_new_item-tran_id
    AND item_no = @ls_new_item-item_no.

    IF sy-subrc = 0.
        " Record exists, just modify status
        ls_dttsit2-prod_stat = ls_new_item-prod_stat.
        ls_dttsit2-trans_stat = ls_new_item-trans_stat.
        MODIFY zmm_sst_dttsit2 FROM ls_dttsit2.
    ELSE.
        " Record doesn't exist, append new line using data from old table with
new status
        MOVE-CORRESPONDING ls_new_item TO ls_dttsit2.
        " Set new PROD_STAT and TRANS_STAT handled by response FORM previously
        INSERT zmm_sst_dttsit2 FROM ls_dttsit2.
    ENDIF.
ENDLOOP.

CATCH cx_ai_application_fault INTO DATA(lx_app_fault).
    PERFORM handle_dtts_application_fault USING lx_app_fault lv_operation CH
ANGING lt_items lt_header.
CATCH cx_ai_system_fault INTO DATA(lx_sys).
    PERFORM handle_api_error_multiple USING lx_sys CHANGING lt_items lt_head
er.
CATCH cx_root INTO DATA(lx_root).
    PERFORM handle_generic_error_multiple USING lx_root CHANGING lt_items lt
_header.
ENDTRY.
ENDIF.
ENDLOOP.

MESSAGE 'Reprocessing Completed.' TYPE 'S'.
ENDMETHOD.

ENDCLASS.

*-----*
* MAIN PROGRAM EXECUTION
*-----*
DATA: go_main TYPE REF TO lcl_dtts_cockpit.

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```

START-OF-SELECTION.
  CREATE OBJECT go_main.
  go_main->get_data( ).
  CALL SCREEN 0100. " Assume a screen 0100 is created to host the ALV container

*&-----*
*& Module STATUS_0100 OUTPUT
*&-----*
MODULE status_0100 OUTPUT.
  SET PF-STATUS 'STANDARD'.
  go_main->display_alv( ).
ENDMODULE.

*&-----*
*& Module USER_COMMAND_0100 INPUT
*&-----*
MODULE user_command_0100 INPUT.
  CASE sy-ucomm.
    WHEN 'BACK' OR 'EXIT' OR 'CANC'.
      LEAVE TO SCREEN 0.
  ENDCASE.
ENDMODULE.

*&-----*
*& SECTION 2: REQUEST PREPARATION FORMS
*&-----*
*&-----*
*&-----*
*& Form ACCEPT_REQUEST
*&-----*
FORM accept_request USING p_mseg_table TYPE table
                        p_header_table TYPE table
                        p_items_table TYPE table
                        CHANGING p_request TYPE zaccept_service_request.

DATA: lwa_request          TYPE zaccept_service_request,
      lwa_product_line     TYPE zaccept_service_request_produ1,
      lv_gtin_formatted    TYPE string,
      lv_quantity_formatted TYPE string,
      lv_batch_formatted   TYPE string,
      lv_exp_date_formatted TYPE string,
      lv_from_gln          TYPE zmm_br_gln,
      lv_auth_gln          TYPE zmm_br_gln.

FIELD-SYMBOLS: <fs_header> TYPE zmm_sst_dtts_hdr,
               <fs_item>   TYPE zmm_sst_dtts_itm.

" Get GLNs from header record
READ TABLE p_header_table ASSIGNING <fs_header> INDEX 1.
IF sy-subrc = 0.
  lv_from_gln = <fs_header>-frm_gln.
  lv_auth_gln = <fs_header>-to_gln.
ENDIF.

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" Loop through ITEMS to build product list
LOOP AT p_items_table ASSIGNING <fs_item>.
*****SKIP unregistered materials*****
  IF <fs_item>-prod_stat EQ 'SUCCESS'.
    CONTINUE.
  ENDIF.

  " Format data from item record (has GTIN, BATCH, EXP_DATE, QUANTITY)
  PERFORM format_data USING <fs_item>-gtin <fs_item>-prod_qty
                           <fs_item>-batch <fs_item>-exp_date
                           CHANGING lv_gtin_formatted lv_quantity_formatted
                                   lv_batch_formatted lv_exp_date_formatted.

  CLEAR lwa_product_line.
  lwa_product_line-gtin      = lv_gtin_formatted.
  lwa_product_line-quantity = lv_quantity_formatted.
  lwa_product_line-bn       = lv_batch_formatted.
  lwa_product_line-xd       = lv_exp_date_formatted.

  APPEND lwa_product_line TO lwa_request-accept_batch_service_request-productlist-pr
oduct.

ENDLOOP.

lwa_request-accept_batch_service_request-fromgln = lv_from_gln.
lwa_request-accept_batch_service_request-authgln = lv_auth_gln.

p_request = lwa_request.

ENDFORM.

*&-----*
*& Form  RETURN_REQUEST
*&-----*
FORM return_request USING p_mseg_table TYPE table
                        p_header_table TYPE table
                        p_items_table TYPE table
                        CHANGING p_request TYPE zreturn_batch_service_request.

DATA: lwa_request          TYPE zreturn_batch_service_request,
      lwa_product_line     TYPE zreturn_batch_service_request3,
      lv_gtin_formatted    TYPE string,
      lv_quantity_formatted TYPE string,
      lv_batch_formatted   TYPE string,
      lv_exp_date_formatted TYPE string,
      lv_to_gln            TYPE zmm_br_gln,
      lv_auth_gln          TYPE zmm_br_gln.

FIELD-SYMBOLS: <fs_header> TYPE zmm_sst_dtts_hdr,
               <fs_item>   TYPE zmm_sst_dtts_itm.

READ TABLE p_header_table ASSIGNING <fs_header> INDEX 1.
IF sy-subrc = 0.
  lv_to_gln = <fs_header>-to_gln.
  lv_auth_gln = <fs_header>-frm_gln.

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ENDIF.

LOOP AT p_items_table ASSIGNING <fs_item>.
*****SKIP unregistered materials*****
  IF <fs_item>-prod_stat EQ 'SUCCESS'.
    CONTINUE.
  ENDIF.

  PERFORM format_data USING <fs_item>-gtin <fs_item>-prod_qty
                           <fs_item>-batch <fs_item>-exp_date
                           CHANGING lv_gtin_formatted lv_quantity_formatted
                                   lv_batch_formatted lv_exp_date_formatted.

  CLEAR lwa_product_line.
  lwa_product_line-gtin      = lv_gtin_formatted.
  lwa_product_line-quantity = lv_quantity_formatted.
  lwa_product_line-bn       = lv_batch_formatted.
  lwa_product_line-xd       = lv_exp_date_formatted.

  APPEND lwa_product_line TO lwa_request-return_batch_service-productlist-product.

ENDLOOP.

lwa_request-return_batch_service-togln = lv_to_gln.
lwa_request-return_batch_service-authgln = lv_auth_gln.

p_request = lwa_request.

ENDFORM.

*&-----*
*& Form  DISPATCH_REQUEST
*&-----*
FORM dispatch_request USING p_mseg_table TYPE table
                           p_header_table TYPE table
                           p_items_table TYPE table
                           CHANGING p_request TYPE zdispatch_batch_service_reque3.

DATA: lwa_request          TYPE zdispatch_batch_service_reque3,
      lwa_product_line     TYPE zdispatch_batch_service_reque1,
      lv_gtin_formatted    TYPE string,
      lv_quantity_formatted TYPE string,
      lv_batch_formatted   TYPE string,
      lv_exp_date_formatted TYPE string,
      lv_to_gln            TYPE zmm_br_gln,
      lv_auth_gln          TYPE zmm_br_gln.

FIELD-SYMBOLS: <fs_header> TYPE zmm_sst_dtts_hdr,
               <fs_item>   TYPE zmm_sst_dtts_itm.

READ TABLE p_header_table ASSIGNING <fs_header> INDEX 1.
IF sy-subrc = 0.
  lv_to_gln = <fs_header>-to_gln.
  lv_auth_gln = <fs_header>-frm_gln.
ENDIF.

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LOOP AT p_items_table ASSIGNING <fs_item>.
*****SKIP unregistered materials*****
  IF <fs_item>-prod_stat EQ 'SUCCESS'.
    CONTINUE.
  ENDIF.
  PERFORM format_data USING <fs_item>-gtin <fs_item>-prod_qty
                           <fs_item>-batch <fs_item>-exp_date
                           CHANGING lv_gtin_formatted lv_quantity_formatted
                                   lv_batch_formatted lv_exp_date_formatted.

  CLEAR lwa_product_line.
  lwa_product_line-gtin      = lv_gtin_formatted.
  lwa_product_line-quantity = lv_quantity_formatted.
  lwa_product_line-bn       = lv_batch_formatted.
  lwa_product_line-xd       = lv_exp_date_formatted.

  APPEND lwa_product_line TO lwa_request-dispatch_batch_service-productlist-product.

ENDLOOP.

lwa_request-dispatch_batch_service-togln = lv_to_gln.
lwa_request-dispatch_batch_service-authgln = lv_auth_gln.

p_request = lwa_request.

ENDFORM.

*&-----*
*& Form  DISPATCH_CANCEL_REQUEST
*&-----*
FORM dispatch_cancel_request USING p_mseg_table TYPE table
                                p_header_table TYPE table
                                p_items_table TYPE table
                                CHANGING p_request TYPE zdispatch_cancel_request.

DATA: lwa_request          TYPE zdispatch_cancel_request,
      lwa_product_line     TYPE zdispatch_cancel_request_prodl,
      lv_gtin_formatted    TYPE string,
      lv_quantity_formatted TYPE string,
      lv_batch_formatted   TYPE string,
      lv_exp_date_formatted TYPE string,
      lv_to_gln            TYPE zmm_br_gln,
      lv_auth_gln          TYPE zmm_br_gln.

FIELD-SYMBOLS: <fs_header> TYPE zmm_sst_dtts_hdr,
               <fs_item>   TYPE zmm_sst_dtts_itm.

READ TABLE p_header_table ASSIGNING <fs_header> INDEX 1.
IF sy-subrc = 0.
  lv_to_gln = <fs_header>-frm_gln.
  lv_auth_gln = <fs_header>-to_gln.
ENDIF.

LOOP AT p_items_table ASSIGNING <fs_item>.

```

```

*****Skip unregistered materials*****
IF <fs_item>-prod_stat EQ 'SUCCESS'.
  CONTINUE.
ENDIF.
PERFORM format_data USING <fs_item>-gtin <fs_item>-prod_qty
                          <fs_item>-batch <fs_item>-exp_date
                          CHANGING lv_gtin_formatted lv_quantity_formatted
                                  lv_batch_formatted lv_exp_date_formatted.

CLEAR lwa_product_line.
lwa_product_line-gtin    = lv_gtin_formatted.
lwa_product_line-quantity = lv_quantity_formatted.
lwa_product_line-bn      = lv_batch_formatted.
lwa_product_line-xd      = lv_exp_date_formatted.

APPEND lwa_product_line TO lwa_request-dispatch_cancel-productlist-product.

ENDLOOP.

lwa_request-dispatch_cancel-togln = lv_to_gln.
lwa_request-dispatch_cancel-authgln = lv_auth_gln.

p_request = lwa_request.

ENDFORM.

*&-----*
*& Form  TRANSFER_REQUEST
*&-----*
FORM transfer_request USING p_mseg_table TYPE table
                          p_header_table TYPE table
                          p_items_table TYPE table
                          CHANGING p_request TYPE ztransfer_batch_service_reque3.

DATA: lwa_request          TYPE ztransfer_batch_service_reque3,
      lwa_product_line     TYPE ztransfer_batch_service_reque2,
      lv_gtin_formatted    TYPE string,
      lv_quantity_formatted TYPE string,
      lv_batch_formatted   TYPE string,
      lv_exp_date_formatted TYPE string,
      lv_to_gln            TYPE zmm_br_gln,
      lv_auth_gln          TYPE zmm_br_gln.

FIELD-SYMBOLS: <fs_header> TYPE zmm_sst_dtts_hdr,
               <fs_item>   TYPE zmm_sst_dtts_itm.

READ TABLE p_header_table ASSIGNING <fs_header> INDEX 1.
IF sy-subrc = 0.
  lv_to_gln = <fs_header>-to_gln.
  lv_auth_gln = <fs_header>-frm_gln.
ENDIF.

LOOP AT p_items_table ASSIGNING <fs_item>.
*****Skip unregistered materials*****
IF <fs_item>-prod_stat EQ 'SUCCESS'.

```

```

        CONTINUE.
    ENDIF.
    PERFORM format_data USING <fs_item>-gtin <fs_item>-prod_qty
                             <fs_item>-batch <fs_item>-exp_date
        CHANGING lv_gtin_formatted lv_quantity_formatted
                 lv_batch_formatted lv_exp_date_formatted.

    CLEAR lwa_product_line.
    lwa_product_line-gtin      = lv_gtin_formatted.
    lwa_product_line-quantity = lv_quantity_formatted.
    lwa_product_line-bn       = lv_batch_formatted.
    lwa_product_line-xd       = lv_exp_date_formatted.

    APPEND lwa_product_line TO lwa_request-transfer_batch_service-productlist-product.

ENDLOOP.

lwa_request-transfer_batch_service-togln = lv_to_gln.
lwa_request-transfer_batch_service-authgln = lv_auth_gln.

p_request = lwa_request.

ENDFORM.

*&-----*
*& Form  TRANSFER_CANCEL_REQUEST
*&-----*
FORM transfer_cancel_request USING p_mseg_table TYPE table
                                p_header_table TYPE table
                                p_items_table TYPE table
                                CHANGING p_request TYPE ztransfer_cancel_request.

DATA: lwa_request          TYPE ztransfer_cancel_request,
      lwa_product_line     TYPE ztransfer_cancel_request_prodl,
      lv_gtin_formatted    TYPE string,
      lv_quantity_formatted TYPE string,
      lv_batch_formatted   TYPE string,
      lv_exp_date_formatted TYPE string,
      lv_to_gln            TYPE zmm_br_gln,
      lv_auth_gln          TYPE zmm_br_gln.

FIELD-SYMBOLS: <fs_header> TYPE zmm_sst_dtts_hdr,
               <fs_item>   TYPE zmm_sst_dtts_itm.

READ TABLE p_header_table ASSIGNING <fs_header> INDEX 1.
IF sy-subrc = 0.
    lv_to_gln = <fs_header>-to_gln.
    lv_auth_gln = <fs_header>-frm_gln.
ENDIF.

LOOP AT p_items_table ASSIGNING <fs_item>.
*****SKIP unregistered materials*****
    IF <fs_item>-prod_stat EQ 'SUCCESS'.
        CONTINUE.
    ENDIF.

```

```

PERFORM format_data USING <fs_item>-gtin <fs_item>-prod_qty
                        <fs_item>-batch <fs_item>-exp_date
CHANGING lv_gtin_formatted lv_quantity_formatted
        lv_batch_formatted lv_exp_date_formatted.

CLEAR lwa_product_line.
lwa_product_line-gtin      = lv_gtin_formatted.
lwa_product_line-quantity = lv_quantity_formatted.
lwa_product_line-bn       = lv_batch_formatted.
lwa_product_line-xd       = lv_exp_date_formatted.

APPEND lwa_product_line TO lwa_request-transfer_cancel-productlist-product.

ENDLOOP.

lwa_request-transfer_cancel-togln = lv_to_gln.
lwa_request-transfer_cancel-authgln = lv_auth_gln.

p_request = lwa_request.

ENDFORM.

*&-----*
*& Form CONSUME_REQUEST
*&-----*
FORM consume_request USING p_mseg_table TYPE table
                        p_header_table TYPE table
                        p_items_table TYPE table
CHANGING p_request TYPE zconsume_service_request.

DATA: lwa_request          TYPE zconsume_service_request,
      lwa_product_line     TYPE zconsume_service_request_prodl,
      lv_gtin_formatted    TYPE string,
      lv_quantity_formatted TYPE string,
      lv_batch_formatted   TYPE string,
      lv_exp_date_formatted TYPE string,
      lv_auth_gln          TYPE zmm_br_gln.

FIELD-SYMBOLS: <fs_header> TYPE zmm_sst_dtts_hdr,
               <fs_item>   TYPE zmm_sst_dtts_itm.

READ TABLE p_header_table ASSIGNING <fs_header> INDEX 1.
IF sy-subrc = 0.
    lv_auth_gln = <fs_header>-frm_gln.
ENDIF.

LOOP AT p_items_table ASSIGNING <fs_item>.
*****SKIP unregistered materials*****
    IF <fs_item>-prod_stat EQ 'SUCCESS'.
        CONTINUE.
    ENDIF.
    PERFORM format_data USING <fs_item>-gtin <fs_item>-prod_qty
                            <fs_item>-batch <fs_item>-exp_date
    CHANGING lv_gtin_formatted lv_quantity_formatted
            lv_batch_formatted lv_exp_date_formatted.

```



```

CLEAR lwa_product_line.
lwa_product_line-gtin = lv_gtin_formatted.
lwa_product_line-sn = <fs_item>-sr_number.
lwa_product_line-bn   = lv_batch_formatted.
lwa_product_line-xd   = lv_exp_date_formatted.

APPEND lwa_product_line TO lwa_request-consume_service-productlist-product.

ENDLOOP.

lwa_request-consume_service-authgln = lv_auth_gln.

p_request = lwa_request.

ENDFORM.

*&-----*
*& Form  CONSUME_CANCEL_REQUEST
*&-----*
FORM consume_cancel_request USING p_mseg_table TYPE table
                                p_header_table TYPE table
                                p_items_table TYPE table
                                CHANGING p_request TYPE zconsume_cancel_request.

DATA: lwa_request          TYPE zconsume_cancel_request,
      lwa_product_line     TYPE zconsume_cancel_request_produ1,
      lv_gtin_formatted    TYPE string,
      lv_quantity_formatted TYPE string,
      lv_batch_formatted   TYPE string,
      lv_exp_date_formatted TYPE string,
      lv_auth_gln          TYPE zmm_br_gln.

FIELD-SYMBOLS: <fs_header> TYPE zmm_sst_dtts_hdr,
               <fs_item>   TYPE zmm_sst_dtts_itm.

READ TABLE p_header_table ASSIGNING <fs_header> INDEX 1.
IF sy-subrc = 0.
    lv_auth_gln = <fs_header>-to_gln.
ENDIF.

LOOP AT p_items_table ASSIGNING <fs_item>.
*****SKIP unregistered materials*****
    IF <fs_item>-prod_stat EQ 'SUCCESS'.
        CONTINUE.
    ENDIF.
    PERFORM format_data USING <fs_item>-gtin <fs_item>-prod_qty
                             <fs_item>-batch <fs_item>-exp_date
                             CHANGING lv_gtin_formatted lv_quantity_formatted
                                     lv_batch_formatted lv_exp_date_formatted.

    CLEAR lwa_product_line.
    lwa_product_line-gtin = lv_gtin_formatted.
    lwa_product_line-sn = <fs_item>-sr_number.
    lwa_product_line-bn   = lv_batch_formatted.

```

```

        lwa_product_line-xd    = lv_exp_date_formatted.

        APPEND lwa_product_line TO lwa_request-consume_cancel-productlist-product.

    ENDLOOP.

    lwa_request-consume_cancel-authgln = lv_auth_gln.

    p_request = lwa_request.

ENDFORM.

*&-----*
*& Form  PHARMACY_SALE_REQUEST
*&-----*
FORM pharmacy_sale_request  USING p_mseg_table TYPE table
                                p_header_table TYPE table
                                p_items_table TYPE table
                                CHANGING p_request TYPE zpharmacy_sale_request.

DATA: lwa_request              TYPE zpharmacy_sale_request,
      lwa_product_line        TYPE zpharmacy_sale_request_producl,
      lv_gtin_formatted       TYPE string,
      lv_quantity_formatted   TYPE string,
      lv_batch_formatted      TYPE string,
      lv_exp_date_formatted   TYPE string,
      lv_to_gln               TYPE zmm_br_gln,
      lv_auth_gln             TYPE zmm_br_gln,
      lv_prescription_date     TYPE string,
      lv_year(4)              TYPE c,
      lv_month(2)             TYPE c,
      lv_day(2)               TYPE c.

FIELD-SYMBOLS: <fs_header> TYPE zmm_sst_dtts_hdr,
               <fs_item>   TYPE zmm_sst_dtts_itm.

READ TABLE p_header_table ASSIGNING <fs_header> INDEX 1.
IF sy-subrc = 0.
    lv_to_gln = <fs_header>-to_gln.
    lv_auth_gln = <fs_header>-frm_gln.
ENDIF.

" Format current date as YYYY-MM-DD for prescription date
lv_year  = sy-datum+0(4).
lv_month = sy-datum+4(2).
lv_day   = sy-datum+6(2).
lv_prescription_date = |{ lv_year }-{ lv_month }-{ lv_day }|.

LOOP AT p_items_table ASSIGNING <fs_item>.
*****SKIP unregistered materials*****
    IF <fs_item>-prod_stat EQ 'SUCCESS'.
        CONTINUE.
    ENDIF.
    PERFORM format_data USING <fs_item>-gtin <fs_item>-prod_qty
                            <fs_item>-batch <fs_item>-exp_date

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```

        CHANGING lv_gtin_formatted lv_quantity_formatted
                lv_batch_formatted lv_exp_date_formatted.

CLEAR lwa_product_line.
lwa_product_line-gtin = lv_gtin_formatted.
lwa_product_line-sn = <fs_item>-sr_number.
lwa_product_line-bn  = lv_batch_formatted.
lwa_product_line-xd  = lv_exp_date_formatted.

APPEND lwa_product_line TO lwa_request-pharmacy_sale-productlist-product.

ENDLOOP.

" Check if toGLN = authGLN (invalid destination)
IF lv_to_gln = lv_auth_gln.
    lwa_request-pharmacy_sale-togln = '00000000000000'.
ELSEIF lv_to_gln IS NOT INITIAL.
    lwa_request-pharmacy_sale-togln = lv_to_gln.
ELSE.
    lwa_request-pharmacy_sale-togln = '00000000000000'.
ENDIF.

lwa_request-pharmacy_sale-authgln = lv_auth_gln.
lwa_request-pharmacy_sale-doctorid = 'NA'.
lwa_request-pharmacy_sale-patientnationalid = 'NA'.
lwa_request-pharmacy_sale-prescriptionid = 'NA'.
lwa_request-pharmacy_sale-prescriptiondate = lv_prescription_date.

p_request = lwa_request.

ENDFORM.

*&-----*
*& Form  PHARMACY_SALE_CANCEL_REQUEST
*&-----*
FORM pharmacy_sale_cancel_request USING p_mseg_table TYPE table
                                         p_header_table TYPE table
                                         p_items_table TYPE table
                                         CHANGING p_request TYPE zpharmacy_sale_cancel_reques
t3.

DATA: lwa_request          TYPE zpharmacy_sale_cancel_request3,
      lwa_product_line     TYPE zpharmacy_sale_cancel_request2,
      lv_gtin_formatted    TYPE string,
      lv_quantity_formatted TYPE string,
      lv_batch_formatted   TYPE string,
      lv_exp_date_formatted TYPE string,
      lv_to_gln            TYPE zmm_br_gln,
      lv_auth_gln          TYPE zmm_br_gln.

FIELD-SYMBOLS: <fs_header> TYPE zmm_sst_dtts_hdr,
               <fs_item>   TYPE zmm_sst_dtts_itm.

READ TABLE p_header_table ASSIGNING <fs_header> INDEX 1.
IF sy-subrc = 0.

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```

lv_to_gln = <fs_header>-to_gln.
lv_auth_gln = <fs_header>-to_gln.
ENDIF.

LOOP AT p_items_table ASSIGNING <fs_item>.
*****SKIP unregistered materials*****
IF <fs_item>-prod_stat EQ 'SUCCESS'.
CONTINUE.
ENDIF.
PERFORM format_data USING <fs_item>-gtin <fs_item>-prod_qty
                        <fs_item>-batch <fs_item>-exp_date
                        CHANGING lv_gtin_formatted lv_quantity_formatted
                        lv_batch_formatted lv_exp_date_formatted.

CLEAR lwa_product_line.
lwa_product_line-gtin = lv_gtin_formatted.
lwa_product_line-sn = <fs_item>-sr_number.
lwa_product_line-bn  = lv_batch_formatted.
lwa_product_line-xd  = lv_exp_date_formatted.

APPEND lwa_product_line TO lwa_request-pharmacy_sale_cancel-productlist-product.

ENDLOOP.

" Check if toGLN = authGLN (invalid destination)
IF lv_to_gln = lv_auth_gln.
lwa_request-pharmacy_sale_cancel-togln = '00000000000000'.
ELSEIF lv_to_gln IS NOT INITIAL.
lwa_request-pharmacy_sale_cancel-togln = lv_to_gln.
ELSE.
lwa_request-pharmacy_sale_cancel-togln = '00000000000000'.
ENDIF.

lwa_request-pharmacy_sale_cancel-authgln = lv_auth_gln.
lwa_request-pharmacy_sale_cancel-prescriptionid = 'NA'.

p_request = lwa_request.

ENDFORM.

*&-----*
*& Form  DEACTIVATE_REQUEST
*&-----*
FORM deactivate_request USING p_mseg_table TYPE table
                        p_header_table TYPE table
                        p_items_table TYPE table
                        CHANGING p_request TYPE zdeactivation_service_request3.

DATA: lwa_request          TYPE zdeactivation_service_request3,
      lwa_product_line     TYPE zdeactivation_service_request1,
      lv_gtin_formatted    TYPE string,
      lv_quantity_formatted TYPE string,
      lv_batch_formatted   TYPE string,
      lv_exp_date_formatted TYPE string,
      lv_auth_gln          TYPE zmm_br_gln,
      lv_grund             TYPE mb_grbew,

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lv_grund_desc          TYPE grtxt.

FIELD-SYMBOLS: <fs_header> TYPE zmm_sst_dtts_hdr,
               <fs_item>   TYPE zmm_sst_dtts_itm,
               <fs_mseg>   TYPE matdoc.

READ TABLE p_header_table ASSIGNING <fs_header> INDEX 1.
IF sy-subrc = 0.
    lv_auth_gln = <fs_header>-frm_gln.
ENDIF.

READ TABLE p_mseg_table ASSIGNING <fs_mseg> INDEX 1.
IF sy-subrc = 0 AND <fs_mseg>-grund IS NOT INITIAL.
    lv_grund = <fs_mseg>-grund.

    " Get description from T157E table for English language
    SELECT SINGLE grtxt FROM t157e WHERE spras = 'E'
        AND bwart = @<fs_mseg>-bwart AND grund = @lv_grund INTO @lv_grund_desc.

    IF sy-subrc <> 0.
        " Fallback if description not found
        lv_grund_desc = 'Deactivation'.
    ENDIF.
ELSE.
    " Default values if GRUND is empty
    lv_grund = '30'.
    lv_grund_desc = 'Damaged Product'.
ENDIF.

LOOP AT p_items_table ASSIGNING <fs_item>.
*****SKip unregistered materials*****
    IF <fs_item>-prod_stat EQ 'SUCCESS'.
        CONTINUE.
    ENDIF.
    PERFORM format_data USING <fs_item>-gtin <fs_item>-prod_qty
                           <fs_item>-batch <fs_item>-exp_date
        CHANGING lv_gtin_formatted lv_quantity_formatted
                 lv_batch_formatted lv_exp_date_formatted.

    CLEAR lwa_product_line.
    lwa_product_line-gtin = lv_gtin_formatted.
    lwa_product_line-sn = <fs_item>-sr_number.
    lwa_product_line-bn  = lv_batch_formatted.
    lwa_product_line-xd  = lv_exp_date_formatted.

    APPEND lwa_product_line TO lwa_request-deactivation_request-productlist-product.

ENDLOOP.
SHIFT lv_grund LEFT DELETING LEADING '0'.
lwa_request-deactivation_request-authgln = lv_auth_gln.
lwa_request-deactivation_request-dr = lv_grund.
lwa_request-deactivation_request-explanation = lv_grund_desc.

p_request = lwa_request.

```

```

ENDFORM.

*&-----*
*& Form  DEACTIVATE_CANCEL_REQUEST
*&-----*
FORM deactivate_cancel_request USING p_mseg_table TYPE table
                                p_header_table TYPE table
                                p_items_table TYPE table
                                CHANGING p_request TYPE zdeactivation_cancel_request.

DATA: lwa_request          TYPE zdeactivation_cancel_request,
      lwa_product_line     TYPE zdeactivation_cancel_request_1,
      lv_gtin_formatted    TYPE string,
      lv_quantity_formatted TYPE string,
      lv_batch_formatted   TYPE string,
      lv_exp_date_formatted TYPE string,
      lv_auth_gln          TYPE zmm_br_gln.

FIELD-SYMBOLS: <fs_header> TYPE zmm_sst_dtts_hdr,
               <fs_item>   TYPE zmm_sst_dtts_itm.

READ TABLE p_header_table ASSIGNING <fs_header> INDEX 1.
IF sy-subrc = 0.
    lv_auth_gln = <fs_header>-to_gln.
ENDIF.

LOOP AT p_items_table ASSIGNING <fs_item>.
*****SKIP unregistered materials*****
    IF <fs_item>-prod_stat EQ 'SUCCESS'.
        CONTINUE.
    ENDIF.
    PERFORM format_data USING <fs_item>-gtin <fs_item>-prod_qty
                             <fs_item>-batch <fs_item>-exp_date
                             CHANGING lv_gtin_formatted lv_quantity_formatted
                                       lv_batch_formatted lv_exp_date_formatted.

    CLEAR lwa_product_line.
    lwa_product_line-gtin = lv_gtin_formatted.
    lwa_product_line-bn   = lv_batch_formatted.
    lwa_product_line-sn   = <fs_item>-sr_number.
    lwa_product_line-xd   = lv_exp_date_formatted.

    APPEND lwa_product_line TO lwa_request-deactivation_cancel-productlist-product.

ENDLOOP.

lwa_request-deactivation_cancel-authgln = lv_auth_gln.

p_request = lwa_request.

ENDFORM.

*&-----*
*& SECTION 3: RESPONSE PROCESSING FORMS
*&-----*
*&-----*

```

```

*&-----*
*& Form  ACCEPT_RESPONSE
*&-----*
FORM accept_response USING p_response TYPE zaccept_service_response
                        CHANGING p_header_table TYPE table
                        p_items_table TYPE table.

DATA: lwa_product_resp TYPE zaccept_service_response_prod1,
      lv_error_code    TYPE char5,
      lv_description    TYPE char255,
      lv_notification_id TYPE string,
      lv_all_success    TYPE abap_bool VALUE abap_true.
DATA: lv_item_gtin TYPE string, lv_resp_gtin TYPE string.

FIELD-SYMBOLS: <fs_header> TYPE zmm_sst_dtts_hdr,
               <fs_item>   TYPE zmm_sst_dtts_itm.

lv_notification_id = p_response-accept_batch_service_response-notificationid.

LOOP AT p_response-accept_batch_service_response-productlist-product
      INTO lwa_product_resp.

  LOOP AT p_items_table ASSIGNING <fs_item>.
    lv_item_gtin = <fs_item>-gtin.
    lv_resp_gtin = lwa_product_resp-gtin.
    SHIFT lv_item_gtin LEFT DELETING LEADING '0'.
    SHIFT lv_resp_gtin LEFT DELETING LEADING '0'.

    IF lv_item_gtin = lv_resp_gtin
    AND <fs_item>-batch = lwa_product_resp-bn
    AND <fs_item>-prod_qty = lwa_product_resp-quantity.
      <fs_item>-notif_id = lv_notification_id.
      <fs_item>-tr_response = lwa_product_resp-rc.
      IF lwa_product_resp-rc = '000000'.
        <fs_item>-prod_stat = 'SUCCESS'.
      ELSE.
        <fs_item>-prod_stat = 'ERROR'.
        lv_all_success = abap_false.
      ENDIF.

      lv_error_code = lwa_product_resp-rc.
      PERFORM get_error_description
        USING lv_error_code
        CHANGING lv_description.

      <fs_item>-trans_stat = lv_description.
      <fs_item>-changed_date = sy-datum.
      <fs_item>-changed_time = sy-uzeit.
      <fs_item>-changed_by = sy-uname.
      EXIT.
    ENDIF.
  ENDLOOP.

ENDLOOP.

```

```

LOOP AT p_header_table ASSIGNING <fs_header>.
  IF lv_all_success = abap_true.
    <fs_header>-status = 'SUCCESS'.
  ELSE.
    <fs_header>-status = 'ERROR'.
  ENDIF.
  <fs_header>-changed_date = sy-datum.
  <fs_header>-changed_time = sy-uzeit.
  <fs_header>-changed_by   = sy-uname.
ENDLOOP.

ENDFORM.

*&-----*
*& Form  RETURN_RESPONSE
*&-----*
FORM return_response USING p_response TYPE zreturn_batch_service_response
                        CHANGING p_header_table TYPE table
                        p_items_table TYPE table.

  DATA: lwa_product_resp  TYPE zreturn_batch_service_respons3,
        lv_error_code     TYPE char5,
        lv_description     TYPE char255,
        lv_notification_id TYPE string,
        lv_all_success     TYPE abap_bool VALUE abap_true.
  DATA: lv_item_gtin TYPE string, lv_resp_gtin TYPE string.
  FIELD-SYMBOLS: <fs_header> TYPE zmm_sst_dtts_hdr,
                <fs_item>   TYPE zmm_sst_dtts_itm.

  lv_notification_id = p_response-return_batch_service_response-notificationid.

  LOOP AT p_response-return_batch_service_response-productlist-product
    INTO lwa_product_resp.

    LOOP AT p_items_table ASSIGNING <fs_item>.
      lv_item_gtin = <fs_item>-gtin.
      lv_resp_gtin = lwa_product_resp-gtin.
      SHIFT lv_item_gtin LEFT DELETING LEADING '0'.
      SHIFT lv_resp_gtin LEFT DELETING LEADING '0'.

      IF lv_item_gtin = lv_resp_gtin
      AND <fs_item>-batch = lwa_product_resp-bn
      AND <fs_item>-prod_qty = lwa_product_resp-quantity.
        <fs_item>-notif_id   = lv_notification_id.
        <fs_item>-tr_response = lwa_product_resp-rc.
        IF lwa_product_resp-rc = '00000'.
          <fs_item>-prod_stat = 'SUCCESS'.
        ELSE.
          <fs_item>-prod_stat = 'ERROR'.
          lv_all_success = abap_false.
        ENDIF.

        lv_error_code = lwa_product_resp-rc.
        PERFORM get_error_description

```



```

        USING lv_error_code
        CHANGING lv_description.

        <fs_item>-trans_stat = lv_description.
        <fs_item>-changed_date = sy-datum.
        <fs_item>-changed_time = sy-uzeit.
        <fs_item>-changed_by   = sy-uname.
        EXIT.
    ENDIF.
ENDLOOP.

ENDLOOP.

LOOP AT p_header_table ASSIGNING <fs_header>.
    IF lv_all_success = abap_true.
        <fs_header>-status = 'SUCCESS'.
    ELSE.
        <fs_header>-status = 'ERROR'.
    ENDIF.
    <fs_header>-changed_date = sy-datum.
    <fs_header>-changed_time = sy-uzeit.
    <fs_header>-changed_by   = sy-uname.
ENDLOOP.

ENDFORM.

*&-----*
*& Form DISPATCH_RESPONSE
*&-----*
FORM dispatch_response USING p_response TYPE zdispatch_batch_service_respo3
                          CHANGING p_header_table TYPE table
                                   p_items_table TYPE table.

    DATA: lwa_product_resp TYPE zdispatch_batch_service_respo1,
           lv_error_code   TYPE char5,
           lv_description   TYPE char255,
           lv_notification_id TYPE string,
           lv_all_success   TYPE abap_bool VALUE abap_true.
    DATA: lv_item_gtin TYPE string, lv_resp_gtin TYPE string.

    FIELD-SYMBOLS: <fs_header> TYPE zmm_sst_dtts_hdr,
                  <fs_item>   TYPE zmm_sst_dtts_itm.

    lv_notification_id = p_response-dispatch_batch_service_respons-notificationid.

    LOOP AT p_response-dispatch_batch_service_respons-productlist-product
        INTO lwa_product_resp.

        LOOP AT p_items_table ASSIGNING <fs_item>.
            lv_item_gtin = <fs_item>-gtin.
            lv_resp_gtin = lwa_product_resp-gtin.
            SHIFT lv_item_gtin LEFT DELETING LEADING '0'.
            SHIFT lv_resp_gtin LEFT DELETING LEADING '0'.

            IF lv_item_gtin = lv_resp_gtin

```

```

AND <fs_item>-batch = lwa_product_resp-bn
AND <fs_item>-prod_qty = lwa_product_resp-quantity.
  <fs_item>-notif_id    = lv_notification_id.
  <fs_item>-tr_response = lwa_product_resp-rc.
  IF lwa_product_resp-rc = '00000'.
    <fs_item>-prod_stat = 'SUCCESS'.
  ELSE.
    <fs_item>-prod_stat = 'ERROR'.
    lv_all_success = abap_false.
  ENDIF.

  lv_error_code = lwa_product_resp-rc.
  PERFORM get_error_description
    USING lv_error_code
    CHANGING lv_description.

  <fs_item>-trans_stat = lv_description.
  <fs_item>-changed_date = sy-datum.
  <fs_item>-changed_time = sy-uzeit.
  <fs_item>-changed_by   = sy-uname.
  EXIT.
ENDIF.
ENDLOOP.

ENDLOOP.

LOOP AT p_header_table ASSIGNING <fs_header>.
  IF lv_all_success = abap_true.
    <fs_header>-status = 'SUCCESS'.
  ELSE.
    <fs_header>-status = 'ERROR'.
  ENDIF.
  <fs_header>-changed_date = sy-datum.
  <fs_header>-changed_time = sy-uzeit.
  <fs_header>-changed_by   = sy-uname.
ENDLOOP.

ENDFORM.

*&-----*
*& Form  DISPATCH_CANCEL_RESPONSE
*&-----*
FORM dispatch_cancel_response USING p_response TYPE zdispatch_cancel_response
                                CHANGING p_header_table TYPE table
                                p_items_table TYPE table.

DATA: lwa_product_resp  TYPE zdispatch_cancel_response_pro1,
      lv_error_code     TYPE char5,
      lv_description    TYPE char255,
      lv_notification_id TYPE string,
      lv_all_success     TYPE abap_bool VALUE abap_true.
DATA: lv_item_gtin TYPE string, lv_resp_gtin TYPE string.
FIELD-SYMBOLS: <fs_header> TYPE zmm_sst_dtts_hdr,
               <fs_item>   TYPE zmm_sst_dtts_itm.

```

```

lv_notification_id = p_response-dispatch_cancel_response-notificationid.

LOOP AT p_response-dispatch_cancel_response-productlist-product
  INTO lwa_product_resp.

  LOOP AT p_items_table ASSIGNING <fs_item>.
    lv_item_gtin = <fs_item>-gtin.
    lv_resp_gtin = lwa_product_resp-gtin.
    SHIFT lv_item_gtin LEFT DELETING LEADING '0'.
    SHIFT lv_resp_gtin LEFT DELETING LEADING '0'.

    IF lv_item_gtin = lv_resp_gtin
      AND <fs_item>-batch = lwa_product_resp-bn
      AND <fs_item>-prod_qty = lwa_product_resp-quantity.
      <fs_item>-notif_id = lv_notification_id.
      <fs_item>-tr_response = lwa_product_resp-rc.
      IF lwa_product_resp-rc = '00000'.
        <fs_item>-prod_stat = 'SUCCESS'.
      ELSE.
        <fs_item>-prod_stat = 'ERROR'.
        lv_all_success = abap_false.
      ENDIF.
    ELSE.
      lv_error_code = lwa_product_resp-rc.
      PERFORM get_error_description
        USING lv_error_code
        CHANGING lv_description.

      <fs_item>-trans_stat = lv_description.
      <fs_item>-changed_date = sy-datum.
      <fs_item>-changed_time = sy-uzeit.
      <fs_item>-changed_by = sy-uname.
      EXIT.
    ENDIF.
  ENDLOOP.

ENDLOOP.

LOOP AT p_header_table ASSIGNING <fs_header>.
  IF lv_all_success = abap_true.
    <fs_header>-status = 'SUCCESS'.
  ELSE.
    <fs_header>-status = 'ERROR'.
  ENDIF.
  <fs_header>-changed_date = sy-datum.
  <fs_header>-changed_time = sy-uzeit.
  <fs_header>-changed_by = sy-uname.
ENDLOOP.

ENDFORM.

*&-----*
*& Form  TRANSFER_RESPONSE
*&-----*
FORM transfer_response USING p_response TYPE ztransfer_batch_service_respo3

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                CHANGING p_header_table TYPE table
                    p_items_table TYPE table.

DATA: lwa_product_resp    TYPE ztransfer_batch_service_respo2,
      lv_error_code       TYPE char5,
      lv_description       TYPE char255,
      lv_notification_id   TYPE string,
      lv_all_success       TYPE abap_bool VALUE abap_true.
DATA: lv_item_gtin TYPE string, lv_resp_gtin TYPE string.
FIELD-SYMBOLS: <fs_header> TYPE zmm_sst_dtts_hdr,
               <fs_item>    TYPE zmm_sst_dtts_itm.

lv_notification_id = p_response-transfer_batch_service_respons-notificationid.

LOOP AT p_response-transfer_batch_service_respons-productlist-product
      INTO lwa_product_resp.

  LOOP AT p_items_table ASSIGNING <fs_item>.
    lv_item_gtin = <fs_item>-gtin.
    lv_resp_gtin = lwa_product_resp-gtin.
    SHIFT lv_item_gtin LEFT DELETING LEADING '0'.
    SHIFT lv_resp_gtin LEFT DELETING LEADING '0'.

    IF lv_item_gtin = lv_resp_gtin
    AND <fs_item>-batch = lwa_product_resp-bn
    AND <fs_item>-prod_qty = lwa_product_resp-quantity.
      <fs_item>-notif_id    = lv_notification_id.
      <fs_item>-tr_response = lwa_product_resp-rc.
      IF lwa_product_resp-rc = '00000'.
        <fs_item>-prod_stat = 'SUCCESS'.
      ELSE.
        <fs_item>-prod_stat = 'ERROR'.
        lv_all_success = abap_false.
      ENDIF.

      lv_error_code = lwa_product_resp-rc.
      PERFORM get_error_description
        USING lv_error_code
        CHANGING lv_description.

      <fs_item>-trans_stat = lv_description.
      <fs_item>-changed_date = sy-datum.
      <fs_item>-changed_time = sy-uzeit.
      <fs_item>-changed_by   = sy-uname.
      EXIT.
    ENDIF.
  ENDLOOP.

ENDLOOP.

LOOP AT p_header_table ASSIGNING <fs_header>.
  IF lv_all_success = abap_true.
    <fs_header>-status = 'SUCCESS'.
  ELSE.
    <fs_header>-status = 'ERROR'.
  
```

```

ENDIF.
<fs_header>-changed_date = sy-datum.
<fs_header>-changed_time = sy-uzeit.
<fs_header>-changed_by   = sy-uname.
ENDLOOP.

ENDFORM.

*&-----*
*& Form  TRANSFER_CANCEL_RESPONSE
*&-----*
FORM transfer_cancel_response USING p_response TYPE ztransfer_cancel_response
                                CHANGING p_header_table TYPE table
                                p_items_table TYPE table.

DATA: lwa_product_resp  TYPE ztransfer_cancel_response_pro1,
      lv_error_code     TYPE char5,
      lv_description    TYPE char255,
      lv_notification_id TYPE string,
      lv_all_success     TYPE abap_bool VALUE abap_true.
DATA: lv_item_gtin TYPE string, lv_resp_gtin TYPE string.
FIELD-SYMBOLS: <fs_header> TYPE zmm_sst_dtts_hdr,
               <fs_item>   TYPE zmm_sst_dtts_itm.

lv_notification_id = p_response-transfer_cancel_response-notificationid.

LOOP AT p_response-transfer_cancel_response-productlist-product
      INTO lwa_product_resp.

  LOOP AT p_items_table ASSIGNING <fs_item>.
    lv_item_gtin = <fs_item>-gtin.
    lv_resp_gtin = lwa_product_resp-gtin.
    SHIFT lv_item_gtin LEFT DELETING LEADING '0'.
    SHIFT lv_resp_gtin LEFT DELETING LEADING '0'.

    IF lv_item_gtin = lv_resp_gtin
      AND <fs_item>-batch = lwa_product_resp-bn
      AND <fs_item>-prod_qty = lwa_product_resp-quantity.
      <fs_item>-notif_id   = lv_notification_id.
      <fs_item>-tr_response = lwa_product_resp-rc.
      IF lwa_product_resp-rc = '000000'.
        <fs_item>-prod_stat = 'SUCCESS'.
      ELSE.
        <fs_item>-prod_stat = 'ERROR'.
        lv_all_success = abap_false.
      ENDIF.

      lv_error_code = lwa_product_resp-rc.
      PERFORM get_error_description
        USING lv_error_code
        CHANGING lv_description.

      <fs_item>-trans_stat = lv_description.
      <fs_item>-changed_date = sy-datum.
      <fs_item>-changed_time = sy-uzeit.

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        <fs_item>-changed_by    = sy-uname.
        EXIT.
    ENDIF.
ENDLOOP.

ENDLOOP.

LOOP AT p_header_table ASSIGNING <fs_header>.
    IF lv_all_success = abap_true.
        <fs_header>-status = 'SUCCESS'.
    ELSE.
        <fs_header>-status = 'ERROR'.
    ENDIF.
    <fs_header>-changed_date = sy-datum.
    <fs_header>-changed_time = sy-uzeit.
    <fs_header>-changed_by    = sy-uname.
ENDLOOP.

ENDFORM.

*&-----*
*& Form  CONSUME_RESPONSE
*&-----*
FORM consume_response USING p_response TYPE zconsume_service_response
                           CHANGING p_header_table TYPE table
                                   p_items_table TYPE table.

    DATA: lwa_product_resp  TYPE zconsume_service_response_pro1,
           lv_error_code     TYPE char5,
           lv_description     TYPE char255,
           lv_notification_id TYPE string,
           lv_all_success     TYPE abap_bool VALUE abap_true.
    DATA: lv_item_gtin TYPE string, lv_resp_gtin TYPE string.
    FIELD-SYMBOLS: <fs_header> TYPE zmm_sst_dtts_hdr,
                  <fs_item>    TYPE zmm_sst_dtts_itm.

    lv_notification_id = p_response-consume_service_response-notificationid.

    LOOP AT p_response-consume_service_response-productlist-product
          INTO lwa_product_resp.

        LOOP AT p_items_table ASSIGNING <fs_item>.
            lv_item_gtin = <fs_item>-gtin.
            lv_resp_gtin = lwa_product_resp-gtin.
            SHIFT lv_item_gtin LEFT DELETING LEADING '0'.
            SHIFT lv_resp_gtin LEFT DELETING LEADING '0'.

            IF lv_item_gtin = lv_resp_gtin AND <fs_item>-batch = lwa_product_resp-bn AND <fs
            _item>-sr_number = lwa_product_resp-sn.
                <fs_item>-notif_id    = lv_notification_id.
                <fs_item>-tr_response = lwa_product_resp-rc.
                IF lwa_product_resp-rc = '000000'.
                    <fs_item>-prod_stat = 'SUCCESS'.
                ELSE.
                    <fs_item>-prod_stat = 'ERROR'.
                ENDIF.
            ENDIF.
        ENDLOOP.
    ENDLOOP.

```

```

        lv_all_success = abap_false.
    ENDIF.

    lv_error_code = lwa_product_resp-rc.
    PERFORM get_error_description
        USING lv_error_code
        CHANGING lv_description.

    <fs_item>-trans_stat = lv_description.
    <fs_item>-changed_date = sy-datum.
    <fs_item>-changed_time = sy-uzeit.
    <fs_item>-changed_by = sy-uname.
    EXIT.
ENDIF.
ENDLOOP.

ENDLOOP.

LOOP AT p_header_table ASSIGNING <fs_header>.
    IF lv_all_success = abap_true.
        <fs_header>-status = 'SUCCESS'.
    ELSE.
        <fs_header>-status = 'ERROR'.
    ENDIF.
    <fs_header>-changed_date = sy-datum.
    <fs_header>-changed_time = sy-uzeit.
    <fs_header>-changed_by = sy-uname.
ENDLOOP.

ENDFORM.

*&-----*
*& Form CONSUME_CANCEL_RESPONSE
*&-----*
FORM consume_cancel_response USING p_response TYPE zconsume_cancel_response
                                CHANGING p_header_table TYPE table
                                p_items_table TYPE table.

DATA: lwa_product_resp TYPE zconsume_cancel_response_prod1,
      lv_error_code TYPE char5,
      lv_description TYPE char255,
      lv_notification_id TYPE string,
      lv_all_success TYPE abap_bool VALUE abap_true.
DATA: lv_item_gtin TYPE string, lv_resp_gtin TYPE string.
FIELD-SYMBOLS: <fs_header> TYPE zmm_sst_dtts_hdr,
               <fs_item> TYPE zmm_sst_dtts_itm.

lv_notification_id = p_response-consume_cancel_service_respons-notificationid.

LOOP AT p_response-consume_cancel_service_respons-productlist-product
    INTO lwa_product_resp.

    LOOP AT p_items_table ASSIGNING <fs_item>.
        lv_item_gtin = <fs_item>-gtin.
        lv_resp_gtin = lwa_product_resp-gtin.

```

```

SHIFT lv_item_gtin LEFT DELETING LEADING '0'.
SHIFT lv_resp_gtin LEFT DELETING LEADING '0'.

IF lv_item_gtin = lv_resp_gtin AND <fs_item>-batch = lwa_product_resp-bn AND <fs
_item>-sr_number = lwa_product_resp-sn.
  <fs_item>-notif_id    = lv_notification_id.
  <fs_item>-tr_response = lwa_product_resp-rc.
  IF lwa_product_resp-rc = '000000'.
    <fs_item>-prod_stat = 'SUCCESS'.
  ELSE.
    <fs_item>-prod_stat = 'ERROR'.
    lv_all_success = abap_false.
  ENDIF.

  lv_error_code = lwa_product_resp-rc.
  PERFORM get_error_description
    USING lv_error_code
    CHANGING lv_description.

  <fs_item>-trans_stat = lv_description.
  <fs_item>-changed_date = sy-datum.
  <fs_item>-changed_time = sy-uzeit.
  <fs_item>-changed_by   = sy-uname.
  EXIT.
ENDIF.
ENDLOOP.

ENDLOOP.

LOOP AT p_header_table ASSIGNING <fs_header>.
  IF lv_all_success = abap_true.
    <fs_header>-status = 'SUCCESS'.
  ELSE.
    <fs_header>-status = 'ERROR'.
  ENDIF.
  <fs_header>-changed_date = sy-datum.
  <fs_header>-changed_time = sy-uzeit.
  <fs_header>-changed_by   = sy-uname.
ENDLOOP.
ENDFORM.

*&-----*
*& Form  PHARMACY_SALE_RESPONSE
*&-----*
FORM pharmacy_sale_response USING p_response TYPE zpharmacy_sale_response
                                CHANGING p_header_table TYPE table
                                p_items_table TYPE table.

DATA: lwa_product_resp  TYPE zpharmacy_sale_response_produ1,
      lv_error_code     TYPE char5,
      lv_description    TYPE char255,
      lv_notification_id TYPE string,
      lv_all_success     TYPE abap_bool VALUE abap_true.
DATA: lv_item_gtin TYPE string, lv_resp_gtin TYPE string.
FIELD-SYMBOLS: <fs_header> TYPE zmm_sst_dtts_hdr,
               <fs_item>   TYPE zmm_sst_dtts_itm.

```



```

lv_notification_id = p_response-pharmacy_sale_service_response-notificationid.

LOOP AT p_response-pharmacy_sale_service_response-productlist-product
  INTO lwa_product_resp.

  LOOP AT p_items_table ASSIGNING <fs_item>.
    lv_item_gtin = <fs_item>-gtin.
    lv_resp_gtin = lwa_product_resp-gtin.
    SHIFT lv_item_gtin LEFT DELETING LEADING '0'.
    SHIFT lv_resp_gtin LEFT DELETING LEADING '0'.

    IF lv_item_gtin = lv_resp_gtin AND <fs_item>-batch = lwa_product_resp-bn AND <fs
_item>-sr_number = lwa_product_resp-sn.
      <fs_item>-notif_id    = lv_notification_id.
      <fs_item>-tr_response = lwa_product_resp-rc.
      IF lwa_product_resp-rc = '00000'.
        <fs_item>-prod_stat = 'SUCCESS'.
      ELSE.
        <fs_item>-prod_stat = 'ERROR'.
        lv_all_success = abap_false.
      ENDIF.
    ELSE.
      lv_error_code = lwa_product_resp-rc.
      PERFORM get_error_description
        USING lv_error_code
        CHANGING lv_description.

      <fs_item>-trans_stat = lv_description.
      <fs_item>-changed_date = sy-datum.
      <fs_item>-changed_time = sy-uzeit.
      <fs_item>-changed_by   = sy-uname.
      EXIT.
    ENDIF.
  ENDLOOP.

ENDLOOP.

LOOP AT p_header_table ASSIGNING <fs_header>.
  IF lv_all_success = abap_true.
    <fs_header>-status = 'SUCCESS'.
  ELSE.
    <fs_header>-status = 'ERROR'.
  ENDIF.
  <fs_header>-changed_date = sy-datum.
  <fs_header>-changed_time = sy-uzeit.
  <fs_header>-changed_by   = sy-uname.
ENDLOOP.

ENDFORM.

*&-----*
*& Form  PHARMACY_SALE_CANCEL_RESPONSE
*&-----*
FORM pharmacy_sale_cancel_response
  USING p_response TYPE zpharmacy_sale_cancel_respons3

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                                CHANGING p_header_table TYPE table
                                p_items_table TYPE table.

DATA: lwa_product_resp  TYPE zpharmacy_sale_cancel_respons2,
      lv_error_code     TYPE char5,
      lv_description    TYPE char255,
      lv_notification_id TYPE string,
      lv_all_success    TYPE abap_bool VALUE abap_true.
DATA: lv_item_gtin TYPE string, lv_resp_gtin TYPE string.
FIELD-SYMBOLS: <fs_header> TYPE zmm_sst_dtts_hdr,
               <fs_item>   TYPE zmm_sst_dtts_itm.

lv_notification_id = p_response-pharmacy_sale_cancel_response-notificationid.

LOOP AT p_response-pharmacy_sale_cancel_response-productlist-product
      INTO lwa_product_resp.

  LOOP AT p_items_table ASSIGNING <fs_item>.
    lv_item_gtin = <fs_item>-gtin.
    lv_resp_gtin = lwa_product_resp-gtin.
    SHIFT lv_item_gtin LEFT DELETING LEADING '0'.
    SHIFT lv_resp_gtin LEFT DELETING LEADING '0'.

    IF lv_item_gtin = lv_resp_gtin AND <fs_item>-batch = lwa_product_resp-bn AND <fs
_item>-sr_number = lwa_product_resp-sn.
      <fs_item>-notif_id    = lv_notification_id.
      <fs_item>-tr_response = lwa_product_resp-rc.
      IF lwa_product_resp-rc = '00000'.
        <fs_item>-prod_stat = 'SUCCESS'.
      ELSE.
        <fs_item>-prod_stat = 'ERROR'.
        lv_all_success = abap_false.
      ENDIF.

      lv_error_code = lwa_product_resp-rc.
      PERFORM get_error_description
        USING lv_error_code
        CHANGING lv_description.

      <fs_item>-trans_stat = lv_description.
      <fs_item>-changed_date = sy-datum.
      <fs_item>-changed_time = sy-uzeit.
      <fs_item>-changed_by   = sy-uname.
      EXIT.
    ENDIF.
  ENDLOOP.

ENDLOOP.

LOOP AT p_header_table ASSIGNING <fs_header>.
  IF lv_all_success = abap_true.
    <fs_header>-status = 'SUCCESS'.
  ELSE.
    <fs_header>-status = 'ERROR'.
  ENDIF.
  <fs_header>-changed_date = sy-datum.

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        <fs_header>-changed_time = sy-uzeit.
        <fs_header>-changed_by   = sy-uname.
    ENDLOOP.

ENDFORM.

*&-----*
*& Form  DEACTIVATE_RESPONSE
*&-----*
FORM deactivate_response USING p_response TYPE zdeactivation_service_respons3
                           CHANGING p_header_table TYPE table
                                   p_items_table TYPE table.

    DATA: lwa_product_resp TYPE zdeactivation_service_respons1,
           lv_error_code    TYPE char5,
           lv_description   TYPE char255,
           lv_notification_id TYPE string,
           lv_all_success   TYPE abap_bool VALUE abap_true.
    DATA: lv_item_gtin TYPE string, lv_resp_gtin TYPE string.
    FIELD-SYMBOLS: <fs_header> TYPE zmm_sst_dtts_hdr,
                  <fs_item>   TYPE zmm_sst_dtts_itm.

    lv_notification_id = p_response-deactivation_response-notificationid.

    LOOP AT p_response-deactivation_response-productlist-product
        INTO lwa_product_resp.

        LOOP AT p_items_table ASSIGNING <fs_item>.
            lv_item_gtin = <fs_item>-gtin.
            lv_resp_gtin = lwa_product_resp-gtin.
            SHIFT lv_item_gtin LEFT DELETING LEADING '0'.
            SHIFT lv_resp_gtin LEFT DELETING LEADING '0'.

            IF lv_item_gtin = lv_resp_gtin AND <fs_item>-batch = lwa_product_resp-bn AND <fs
            _item>-sr_number = lwa_product_resp-sn.
                <fs_item>-notif_id    = lv_notification_id.
                <fs_item>-tr_response = lwa_product_resp-rc.
                IF lwa_product_resp-rc = '000000'.
                    <fs_item>-prod_stat = 'SUCCESS'.
                ELSE.
                    <fs_item>-prod_stat = 'ERROR'.
                    lv_all_success = abap_false.
                ENDIF.

                lv_error_code = lwa_product_resp-rc.
                PERFORM get_error_description
                    USING lv_error_code
                    CHANGING lv_description.

                <fs_item>-trans_stat = lv_description.
                <fs_item>-changed_date = sy-datum.
                <fs_item>-changed_time = sy-uzeit.
                <fs_item>-changed_by   = sy-uname.
                EXIT.
            ENDIF.
        ENDLOOP.
    ENDLOOP.

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        ENDLOOP.

    ENDLOOP.

    LOOP AT p_header_table ASSIGNING <fs_header>.
        IF lv_all_success = abap_true.
            <fs_header>-status = 'SUCCESS'.
        ELSE.
            <fs_header>-status = 'ERROR'.
        ENDIF.
        <fs_header>-changed_date = sy-datum.
        <fs_header>-changed_time = sy-uzeit.
        <fs_header>-changed_by   = sy-uname.
    ENDLOOP.

ENDFORM.

*&-----*
*& Form  DEACTIVATE_CANCEL_RESPONSE
*&-----*
FORM deactivate_cancel_response USING p_response TYPE zdeactivation_cancel_response3
                                CHANGING p_header_table TYPE table
                                p_items_table TYPE table.

    DATA: lwa_product_resp  TYPE zdeactivation_cancel_response1,
           lv_error_code     TYPE char5,
           lv_description     TYPE char255,
           lv_notification_id TYPE string,
           lv_all_success     TYPE abap_bool VALUE abap_true.
    DATA: lv_item_gtin TYPE string, lv_resp_gtin TYPE string.
    FIELD-SYMBOLS: <fs_header> TYPE zmm_sst_dtts_hdr,
                  <fs_item>   TYPE zmm_sst_dtts_itm.

    lv_notification_id = p_response-deactivation_cancel_response-notificationid.

    LOOP AT p_response-deactivation_cancel_response-productlist-product
        INTO lwa_product_resp.

        LOOP AT p_items_table ASSIGNING <fs_item>.
            lv_item_gtin = <fs_item>-gtin.
            lv_resp_gtin = lwa_product_resp-gtin.
            SHIFT lv_item_gtin LEFT DELETING LEADING '0'.
            SHIFT lv_resp_gtin LEFT DELETING LEADING '0'.

            IF lv_item_gtin = lv_resp_gtin AND <fs_item>-batch = lwa_product_resp-bn AND <fs
_item>-sr_number = lwa_product_resp-sn.
                <fs_item>-notif_id   = lv_notification_id.
                <fs_item>-tr_response = lwa_product_resp-rc.
                IF lwa_product_resp-rc = '00000'.
                    <fs_item>-prod_stat = 'SUCCESS'.
                ELSE.
                    <fs_item>-prod_stat = 'ERROR'.
                    lv_all_success = abap_false.
                ENDIF.
            ENDIF.
        ENDLOOP.
    ENDLOOP.

```

```

        lv_error_code = lwa_product_resp-rc.
        PERFORM get_error_description
            USING lv_error_code
            CHANGING lv_description.

        <fs_item>-trans_stat = lv_description.
        <fs_item>-changed_date = sy-datum.
        <fs_item>-changed_time = sy-uzeit.
        <fs_item>-changed_by   = sy-uname.
        EXIT.
    ENDIF.
ENDLOOP.

ENDLOOP.

LOOP AT p_header_table ASSIGNING <fs_header>.
    IF lv_all_success = abap_true.
        <fs_header>-status = 'SUCCESS'.
    ELSE.
        <fs_header>-status = 'ERROR'.
    ENDIF.
    <fs_header>-changed_date = sy-datum.
    <fs_header>-changed_time = sy-uzeit.
    <fs_header>-changed_by   = sy-uname.
ENDLOOP.

ENDFORM.
*&-----*
*& SECTION 4: EXCEPTION HANDLERS & SUPPORTING PERFORMS
*&-----*
*&-----*
*&-----*
*& Form  HANDLE_DTTS_APPLICATION_FAULT
*&-----*
*   Generic handler using config table fault attribute
FORM handle_dtts_application_fault USING p_exception TYPE REF TO cx_ai_application_fau
lt
                                p_operation TYPE char20
                                CHANGING p_items_table TYPE table
                                p_header_table TYPE table.

DATA: lv_response_code TYPE string,
      lv_message       TYPE string.

FIELD-SYMBOLS: <fs_item>   TYPE zmm_sst_dtts_itm,
               <fs_header> TYPE zmm_sst_dtts_hdr,
               <fs_field>  TYPE any.

" ===== Extract Fault Details (Direct Access - Flat Structure) =====

TRY.
    " Try to access RESPONSE_CODE directly
    ASSIGN p_exception->'RESPONSE_CODE' TO <fs_field>.
    IF sy-subrc = 0 AND <fs_field> IS ASSIGNED.
        lv_response_code = <fs_field>.
    ENDIF.

```

```

    " Try to access MESSAGE directly
    UNASSIGN <fs_field>.
    ASSIGN p_exception->'MESSAGE' TO <fs_field>.
    IF sy-subrc = 0 AND <fs_field> IS ASSIGNED.
        lv_message = <fs_field>.
    ENDIF.

    CATCH cx_root.
        " Continue to fallback
    ENDTRY.

    " ===== Fallback if extraction failed =====

    IF lv_response_code IS INITIAL.
        lv_response_code = 'APP_FAULT'.
    ENDIF.

    IF lv_message IS INITIAL.
        lv_message = p_exception->get_text( ).
        IF lv_message IS INITIAL.
            lv_message = |DTTS Application Fault - Operation: { p_operation }|.
        ENDIF.
    ENDIF.

    " ===== Update ALL Items with Fault Details =====

    LOOP AT p_items_table ASSIGNING <fs_item>.
        <fs_item>-tr_response = lv_response_code.
        <fs_item>-trans_stat = lv_message.
        <fs_item>-prod_stat = 'ERROR'.
        <fs_item>-notif_id = ''.

        " Update item timestamp
        <fs_item>-changed_date = sy-datum.
        <fs_item>-changed_time = sy-uzeit.
        <fs_item>-changed_by = sy-uname.
    ENDLOOP.

    " Update headers with timestamp and STATUS
    LOOP AT p_header_table ASSIGNING <fs_header>.
        <fs_header>-status = 'ERROR'.
        <fs_header>-changed_date = sy-datum.
        <fs_header>-changed_time = sy-uzeit.
        <fs_header>-changed_by = sy-uname.
    ENDLOOP.

    " Save to database
    IF p_items_table IS NOT INITIAL.
        MODIFY zmm_sst_dtts_itm FROM TABLE p_items_table.
    ENDIF.

    IF p_header_table IS NOT INITIAL.
        MODIFY zmm_sst_dtts_hdr FROM TABLE p_header_table.
    ENDIF.

```

ENDFORM.

```
*&-----*
*& Form  HANDLE_API_ERROR_MULTIPLE
*&-----*
FORM handle_api_error_multiple USING p_exception TYPE REF TO cx_ai_system_fault
                                CHANGING p_items_table TYPE table
                                        p_header_table TYPE table.

DATA: lv_sys_error  TYPE string,
      lv_mpl_id     TYPE string,
      lv_error_code TYPE string.

FIELD-SYMBOLS: <fs_item>  TYPE zmm_sst_dtts_itm,
               <fs_header> TYPE zmm_sst_dtts_hdr.

" Get error message
lv_sys_error = p_exception->get_text( ).

" Extract MPL ID if available
IF lv_sys_error CS 'MPL ID'.
    FIND REGEX 'MPL ID ([A-Za-z0-9_]+)' IN lv_sys_error
        SUBMATCHES lv_mpl_id.
ENDIF.

" Extract DTTS error code if available (e.g., 98765)
IF lv_sys_error CS '<FC>'.
    FIND REGEX '<FC>(\d+)</FC>' IN lv_sys_error
        SUBMATCHES lv_error_code.
ENDIF.

" Update ALL items with error code
LOOP AT p_items_table ASSIGNING <fs_item>.
    <fs_item>-tr_response = lv_error_code.
    <fs_item>-prod_stat = 'ERROR'.
    <fs_item>-trans_stat = lv_sys_error.
    " Update item timestamp
    <fs_item>-changed_date = sy-datum.
    <fs_item>-changed_time = sy-uzeit.
    <fs_item>-changed_by   = sy-uname.
ENDLOOP.

" Update headers
LOOP AT p_header_table ASSIGNING <fs_header>.
    <fs_header>-status = 'ERROR'.
    <fs_header>-changed_date = sy-datum.
    <fs_header>-changed_time = sy-uzeit.
    <fs_header>-changed_by   = sy-uname.
ENDLOOP.

" Save error records to database
MODIFY zmm_sst_dtts_itm FROM TABLE p_items_table.
MODIFY zmm_sst_dtts_hdr FROM TABLE p_header_table.
```

```

ENDFORM.

*&-----*
*& Form  HANDLE_GENERIC_ERROR_MULTIPLE
*&-----*
FORM handle_generic_error_multiple USING p_exception TYPE REF TO cx_root
                                     CHANGING p_items_table TYPE table
                                             p_header_table TYPE table.

DATA: lv_root_error TYPE string.

FIELD-SYMBOLS: <fs_item>   TYPE zmm_sst_dtts_itm,
               <fs_header> TYPE zmm_sst_dtts_hdr.

" Get error message
lv_root_error = p_exception->get_text( ).

" Update ALL items with generic error
LOOP AT p_items_table ASSIGNING <fs_item>.
    <fs_item>-tr_response = 'ERROR'.
    <fs_item>-prod_stat = 'ERROR'.

    " Update item timestamp
    <fs_item>-changed_date = sy-datum.
    <fs_item>-changed_time = sy-uzeit.
    <fs_item>-changed_by   = sy-uname.
ENDLOOP.

" Update headers
LOOP AT p_header_table ASSIGNING <fs_header>.
    <fs_header>-status = 'ERROR'.
    <fs_header>-changed_date = sy-datum.
    <fs_header>-changed_time = sy-uzeit.
    <fs_header>-changed_by   = sy-uname.
ENDLOOP.

" Save ALL records
MODIFY zmm_sst_dtts_itm FROM TABLE p_items_table.
MODIFY zmm_sst_dtts_hdr FROM TABLE p_header_table.

ENDFORM.

*&-----*
*& Form  FORMAT_DATA
*&-----*
FORM format_data USING p_gtin_in TYPE any
                      p_quantity_in TYPE any
                      p_batch_in TYPE any
                      p_exp_date_in TYPE any
                      CHANGING p_gtin_out TYPE string
                              p_quantity_out TYPE string
                              p_batch_out TYPE string
                              p_exp_date_out TYPE string.

```



```

DATA: lv_len      TYPE i,
      lv_qty_int  TYPE string,
      lv_qty_dec  TYPE string,
      lv_year(4)  TYPE c,
      lv_month(2) TYPE c,
      lv_day(2)   TYPE c.

" Format GTIN
p_gtin_out = p_gtin_in.
CONDENSE p_gtin_out NO-GAPS.
lv_len = strlen( p_gtin_out ).
IF lv_len < 14.
    p_gtin_out = |{ p_gtin_out WIDTH = 14 PAD = '0' ALIGN = RIGHT }|.
ENDIF.

" Format Quantity
p_quantity_out = p_quantity_in.
IF p_quantity_out CS '.'.
    SPLIT p_quantity_out AT '.' INTO lv_qty_int lv_qty_dec.
    p_quantity_out = lv_qty_int.
ENDIF.
p_quantity_out = |{ p_quantity_out ALPHA = OUT }|.
CONDENSE p_quantity_out NO-GAPS.

" Format Batch
p_batch_out = p_batch_in.
CONDENSE p_batch_out NO-GAPS.

" Format Expiry Date
CLEAR p_exp_date_out.
IF p_exp_date_in IS NOT INITIAL.
    lv_year  = p_exp_date_in+0(4).
    lv_month = p_exp_date_in+4(2).
    lv_day   = p_exp_date_in+6(2).
    p_exp_date_out = |{ lv_year }-{ lv_month }-{ lv_day }|.
ENDIF.

ENDFORM.
*&-----*
*& Form  GET_ERROR_DESCRIPTION
*&-----*
FORM get_error_description USING p_error_code TYPE char5
                              CHANGING p_description TYPE char255.

CLEAR p_description.

SELECT SINGLE description FROM zdtts_errorlist INTO @p_description
  WHERE error_code = @p_error_code
    AND language = 'en'.

IF sy-subrc <> 0.
    p_description = |Error Code: { p_error_code }|.
ENDIF.

ENDFORM.

```